

Mike Hall

Senior / Lead Ruby on Rails Developer (Contract / High-Velocity IC)

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Senior Ruby on Rails contractor with 17+ years of production Rails and PostgreSQL expertise. Delivers immediate, zero-ramp-up business value for legacy upgrades, complex state machine refactorings, database optimization, asynchronous background processing, and third-party API integrations with strict TDD/RSpec discipline.

EXPERIENCE & LEADERSHIP

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Target Focus: Architected and delivered the Speedfunds instant loan disbursement pipeline (funding to debit cards in minutes) and refactored multi-step Rails workflow state machines, eliminating lateral state corruptions across high-volume customer prequalification pipelines.

Originations IC Delivery & Team Realignment: Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping the Speedfunds instant loan disbursement pipeline (funding to debit cards in minutes vs multi-day ACH). Appointed Software Architect for the Acquisition Lane to resolve systemic fragility, consolidating two attrition-impacted teams into a unified, high-velocity delivery unit before converting to Associate Director, Staff Engineer upon corporate architecture track restructuring.

ACQ Enablement & Architecture Discovery: Founded and technically led the ACQ Enablement team to establish end-to-end system mapping across seven heterogeneous acquisition channels spanning direct-to-consumer web/mobile funnels, direct mail campaigns, partner affiliate APIs, and authenticated renewal workflows. Architected an automated 5-phase PII Remediation deletion engine and data migration across 30+ tables, cataloging legacy clarity_orphan data, eliminating multi-service state machine corruptions, and embedding automated compliance into production Rails code.

Enterprise Resilience & DynamoDB Session Remediation: Architected and delivered the platform migration from client-side cookie storage to server-side DynamoDB session storage across all Rails applications, diagnosing and eliminating a critical CookieOverflow defect that silently dropped 4% of digital loan applications during late-stage offer selection and e-signing; coordinated cross-lane blue/green deployment with zero downtime and zero incidents.

Enterprise Trace & Operational Alignment: Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends. Partnered directly with Cybersecurity, the Enterprise Monitoring Center (EMC), SRE, and Incident Command to align immediate operational monitoring with long-term distributed tracing across end-to-end multi-tier lending workflows.

Sustained Community Enablement & SRE Handoff: Drove a dedicated 3-year cultural and technical enablement initiative: founded Geekfest@OMF (weekly recorded technical brown-bag sessions across the enterprise for 1 year) before evolving the forum into the weekly OpenTelemetry Working Group (OTel WG), scaling voluntary attendance to 40+ cross-lane engineers. Mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable, self-sustaining community of practice while returning to Acquisition Lane architecture representation.

Applied AI & Innovation Leadership: Placed in two corporate hackathons (building early conversational agents with Rasa and automated schema inference tools with Bonsai Buckaroos). Introduced local LLM orchestration and privacy-conscious AI developer workflows at Geekfest@OMF, establishing early evidence-based patterns for legacy code discovery and developer productivity.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Target Focus: Sequenced zero-downtime Ruby and Rails version upgrades, optimized MySQL full-text search indices, and built batch Sidekiq/SendGrid delivery pipelines.

Search Modernization & Infrastructure De-Risking: Eliminated external cluster failure modes and lowered operating footprint by transitioning legacy Sphinx infrastructure to optimized MySQL full-text search and relational indices under live production traffic.

Zero-Downtime Platform Upgrades: Led Ruby and Rails framework upgrades across multiple core applications, sequencing database migrations and dependency updates to maintain uptime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Target Focus: Introduced testable query patterns and caching into a tightly coupled PHP/MySQL CRM backend and extended the Ember.js frontend test suite across contact management workflows.

Testable Query Patterns: Replaced ad-hoc global data access functions with structured, cacheable query patterns in the PHP backend, improving performance and enabling reliable automated testing where none existed before.

Legacy Boundary Isolation: Identified and isolated structural constraints in the core contact management subsystem, reducing risk for incremental change without requiring large-scale rewrites.

Frontend Test Hardening & Boundary Verification: Hardened Ember.js frontend test coverage across core contact management workflows, creating verification gates that allowed safe refactoring of legacy PHP data paths without regression.

Reusable Pagination Constructs: Refactored shared Ember.js pagination and interaction logic into reusable mixins, eliminating duplication across contact list views.

Senior Software Developer

Upcity | Remote | October 2013 - February 2014

Target Focus: Integrated Chargify recurring billing and standardized multi-VM local development environments via Vagrant.

Integrated the Chargify subscription payment platform, launching self-service recurring billing and account management.

Standardized local development environments using multi-VM Vagrant orchestration, eliminating developer environment drift against production.

Senior Software Developer

Viewpoints | Chicago, IL | May 2013 - October 2013

Target Focus: Implemented custom tracking analytics and automated staging/production deployment pipelines for high-traffic consumer web products.

Implemented custom tracking and analytics tools, enabling targeted monetization and direct platform revenue.

Standardized automated testing and staging environments, establishing reliable continuous integration across the engineering team.

Streamlined AWS deployment pipelines and server images, eliminating deployment downtime during peak consumer traffic spikes.

SELECTED PRODUCTION PROJECTS

Principal Architect

WWWorkRemote | Independent Project | 2021 - Present

Rails 8 platform for multi-source job ingestion, semantic matching, and application automation, built as a working laboratory for local-first LLM orchestration and treating scraped third-party text as hostile input.

Data Ingestion & Normalization: Built a multi-source Rails ingestion pipeline and PostgreSQL storage schema, unifying fragmented job market datasets for structured downstream analysis.

Semantic Search & Matching: Implemented vector search embeddings via pgvector, enabling high-precision semantic matching between complex candidate profiles and job requirements.

Local-First AI Orchestration: Designed a provider-agnostic LLM orchestration layer behind a YAML-driven model registry, local-first by default (llama.cpp/Ollama primary, hosted Claude as fallback), so routine inference never leaves the machine while a stronger hosted model can be selected per call site.

Prompt Injection Defense: Built a four-stage guardrails pipeline (normalize, weighted heuristic scan, risk classify, output validate) screening every untrusted scraped document before it reaches a model, scoring known injection patterns plus an instruction-density heuristic to secure ingestion against hostile inputs.

Browser Extension & Lifecycle Capture: Built an MV3 Chrome extension with 16 provider adapters that extracts postings from live ATS pages and captures the full application lifecycle (questions asked and answers given) back into the platform.

Static Signal Integration: Injected static analysis rules and architectural constraints into model context windows, preventing drift between generated outputs and system security contracts.

Principal Architect

Phalanx Duel | Independent Project | 2022 - Present

Designing and building a real-time tactical game platform as a current hands-on laboratory for deterministic systems, product architecture, and controlled AI-assisted engineering.

Designed a deterministic engine with replayable action logs and server-authoritative state transitions, making complex outcomes reproducible and independently verifiable.

Built replay validation, adversarial coverage, and CI verification gates so state transitions can be checked across the complete lifecycle.

Uses controlled agent workflows with bounded tasks, acceptance criteria, automated checks, and human-reviewed outputs rather than treating model output as authority.

Principal Architect

Local AI Orchestration & Developer Runtime | Independent Project | 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

Bidirectional Knowledge Server: Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.

Telemetry-Driven Diagnosis: Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.

Context as a Budgeted Resource: Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

CORE SKILLS & COMPETENCIES

01 Ruby on Rails (Full-Lifecycle 2.x through 8.x)	02 PostgreSQL & MySQL Query Optimization & Indexing	03 Asynchronous Processing (Sidekiq, Resque, Redis)	04 State Machine Refactoring & Data Integrity
05 Third-Party API & Webhook Integrations	06 Test-Driven Development (RSpec, Capybara, FactoryBot)	07 Docker, Linux, Git & Zero-Downtime Deployment	08 Fast Ramp-Up & Autonomous IC Execution

ADDITIONAL EXPERIENCE

- Senior Software Developer, ReachLocal | March 2015 - November 2016 | Remote: Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- Senior Software Developer, BenchPrep | March 2017 - February 2018 | Chicago, IL: Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.
- Open-Source Transition Lead, Coderwall | January 2014 - December 2014 | Remote: Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- Principal Consultant, Tandem | August 2018 - August 2018 | Chicago, IL: Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.

CAREER FOUNDATIONS

1999 - 2011

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that still shape my leadership.

Revenue-critical commerce: Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.

Consulting and software craftsmanship: Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.

Enterprise and operational systems: Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.